

Indian Institute Of Information Technology Guwahati



Project Report

Group Number - 14

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Section: CS21

Branch: B.Tech 2nd Year CSE

Course: DBMS Lab

Code: CS241

Project Delivery

We developed a full-stack web application with a React-based frontend, a Spring Boot backend, and a MySQL database, designed to support four distinct roles.

- **Admin** – The application includes a fully integrated Admin Panel that provides real-time insights through comprehensive statistics and analytics. It also features a dedicated Custom Query Module, enabling administrators to execute database queries directly for advanced data retrieval and management. Additionally, the Admin has control over user access, system configuration, and overall platform monitoring.
- **Office** – The Office Panel serves as a centralized unit to manage properties and agents within a specific city. Currently, five offices operate across five cities, each with restricted access to data relevant only to their registered properties and agents. Offices can oversee agent performance, manage regional listings, and ensure smooth coordination of transactions within their jurisdiction
- **Agent** – The Agent Panel provides agents with tools to manage their holdings, accept new properties from advertised listings, handle client interactions, and perform property transfers. It also allows agents to track ongoing deals, maintain client records, and streamline communication between buyers and sellers. He has a dedicated Holdings panel, from where he can look for his accepted holdings and leave them
- **User** – The User Panel enables users to browse properties, view detailed listings, and interact with the platform to buy and sell properties. Users can also manage their property portfolio, and receive updates on listing status and activity. The user can add new properties in his portfolio and then list them, after this his listing is automatically advertised to the agents, and they can accept the request. User can list 1 property, at most for 2 roles i.e. Rent or Sell.

Individual Contribution

1. Ansik Singh Tomar (2401037)

- Architected the React frontend while maintaining the and global states, user type based access to multiple pages.
- Integrated JWT-based authentication flow with silent refresh token logic, to the frontend
- Integrated the APIs developed by the backend team with the react components.
- Developed the dynamic Admin and Agent dashboards with real-time analytics visualization.

2. Anmol Kumar (2401032)

- Designed the complete backend system architecture and developed scalable REST APIs using Spring Boot following layered architecture
- Implemented secure authentication and authorization using JWT with role-based access control for User, Admin, Office, and Agent modules
- Applied Strategy Design Pattern to build flexible API architecture, enabling easier frontend integration and extensibility
- Developed analytics and revenue tracking system with optimized and complex database queries for efficient data processing and reporting.

3. Anuvud Pratap Singh (2401041)

- Architected a robust relational database schema for properties, listings, agents, and transactions
- Engineered RESTful APIs for property, listing token, and image management with a focus on scalability
- Built and maintained data scraping workflows ensuring consistency, normalization, and reliability of data
- Enhanced query performance by optimizing complex SQL operations used in analytics and query tools

4. Anmol Kumar (2401031)

- Designed intuitive and user-friendly interfaces with a focus on usability and accessibility
- Created wireframes, prototypes, and user flows to enhance user experience across the platform
- Collaborated with developers to ensure seamless implementation of design systems
- Conducted user research and usability testing to improve interaction and visual design.

Opensource Libraries

We used some libraries like, Shad.cn, Aceternity.ui, radix.ui, to make the react components interactive

AI Contribution

In the development of Urban Enclaves, AI models (Gemini 3 Flash and ChatGPT) were utilized as collaborative agents. The integration of AI focused on three core pillars:

1. **UI/UX Refinement** – Generating complex Tailwind CSS configurations and refining Framer Motion animations for smooth transitions

Example Prompt -

“Refactor this React component to use a glassmorphism effect with a blurred background and semi-transparent borders. Use Tailwind CSS and ensure the dark theme maintains high contrast for accessibility”

2. **Logic Debugging** – Debugging JavaScript logic and optimizing React hooks to prevent unnecessary re-renders, and fetch data efficiently from the backend.

Example Prompt -

“I have my Spring Boot backend running on port 8080 and a React frontend on port 5173. My login API call is being blocked by some CORS error, what is it. this is the error message - Access to fetch at 'http://your-live-backend-link:8080/auth/login' from origin 'http://localhost:5173' has been blocked by CORS policy:”

2. Technical Know-How and Research

Example Prompt -

“Instead of the native Fetch AP, I want to use the Axios for API calling in a React project. Which is better for a real estate platform like Urban Enclaves that requires complex error handling. give me a code snippet for a standard GET request with the try catch block”